Physics in Medicine & Biology

PAPER

Power absorption and temperature rise in deep learning based head models for local radiofrequency exposures

To cite this article: Sachiko Kodera *et al* 2025 *Phys. Med. Biol.* **70** 065013

View the article online for updates and enhancements.

You may also like

- A fast and robust algorithm for general defocusing particle tracking
Massimiliano Rossi and Rune Barnkob
- CAM-QUS guided self-tuning modular CNNs with multi-loss functions for fully automated breast lesion classification in ultrasound images
Jarin Tasnim and Md Kamrul Hasan
- Atlas-based automatic planning and 3D–2D fluoroscopic guidance in pelvic trauma surgery
R Han, A Uneri, T De Silva et al.

This content was downloaded from IP address 133.68.125.30 on 11/03/2025 at 14:10

IOP Publishing Phys. Med. Biol. 70 (2025) 065013 https://doi.org/10.1088/1361-6560/adb935

Physics in Medicine & Biology

IPEM Institute of Physics and Engineering in Medicine

CrossMark

RECEIVED
26 December 2024

REVISED
9 February 2025

ACCEPTED FOR PUBLICATION
21 February 2025

PUBLISHED
11 March 2025

PAPER

Power absorption and temperature rise in deep learning based head models for local radiofrequency exposures

Sachiko Kodera[1,*], **Reina Yoshida**[1], **Essam A Rashed**[2,3], **Yinliang Diao**[4], **Hiroyuki Takizawa**[5] and **Akimasa Hirata**[1]

[1] Department of Electrical and Mechanical Engineering, Nagoya Institute of Technology, Nagoya 466-8555, Japan
[2] Graduate School of Information Science, University of Hyogo, Kobe 650-0047, Japan
[3] Advanced Medical Engineering Research Institute, University of Hyogo, Himeji 670-0836, Japan
[4] College of Electronic Engineering, South China Agricultural University, Guangzhou 510642, People's Republic of China
[5] Cyberscience Center, Tohoku University, Sendai 980-8578, Japan
[*] Author to whom any correspondence should be addressed.

E-mail: kodera@nitech.ac.jp

Keywords: local exposure, deep learning, international guidelines, tissue thermal parameter, power absorption, temperature rise

Abstract

Objective. Computational uncertainty and variability of power absorption and temperature rise in humans for radiofrequency (RF) exposure is a critical factor in ensuring human protection. This aspect has been emphasized as a priority. However, accurately modeling head tissue composition and assigning tissue dielectric and thermal properties remains a challenging task. This study investigated the impact of segmentation-based versus segmentation-free models for assessing localized RF exposure. *Approach.* Two computational head models were compared: one employing traditional tissue segmentation and the other leveraging deep learning to estimate tissue dielectric and thermal properties directly from magnetic resonance images. The finite-difference time-domain method and the bioheat transfer equation was solved to assess temperature rise for local exposure. Inter-subject variability and dosimetric uncertainties were analyzed across multiple frequencies. *Main results.* The comparison between the two methods for head modeling demonstrated strong consistency, with differences in peak temperature rise of 7.6 ± 6.4%. The segmentation-free model showed reduced inter-subject variability, particularly at higher frequencies where superficial heating dominates. The maximum relative standard deviation in the inter-subject variability of heating factor was 15.0% at 3 GHz and decreased with increasing frequencies. *Significance.* This study highlights the advantages of segmentation-free deep-learning models for RF dosimetry, particularly in reducing inter-subject variability and improving computational efficiency. While the differences between the two models are relatively small compared to overall dosimetric uncertainty, segmentation-free models offer a promising approach for refining individual-specific exposure assessments. These findings contribute to improving the accuracy and consistency of human protection guidelines against RF electromagnetic field exposure.

1. Introduction

Two international standardization bodies, the international commission on non-ionizing radiation protection (ICNIRP) and the IEEE International Committee on Electromagnetic Safety (ICES), have issued the guidelines (ICNIRP 2010, 2020a) and the standard (IEEE C95.1 2019) to protect against electromagnetic fields <300 GHz (hereafter referred simply as 'guidelines'). For continuous wave exposure at frequencies <100 kHz, the dominant adverse effect is nervous stimulation, whereas heating becomes dominant at >100 kHz. The guidelines establish exposure limits by first deriving or estimating thresholds for adverse effects. After applying a safety factor (ICNIRP 2020b), they established the limits in terms of induced physical quantity, known as the basic restrictions or dosimetric reference limits. The differences between

© 2025 Institute of Physics and Engineering in Medicine. All rights, including for text and data mining, AI training, and similar technologies, are reserved.

these guidelines and the changes from the previous version have been extensively reviewed in (Ziskin *et al* 2018, Hirata *et al* 2019, Hirata and Kodera 2020).

For localized radiofrequency (RF) exposure, local heating is the dominant adverse health effect at excessive exposure levels. Substantial efforts have been made to correlate temperature rise with spatial average specific absorption rate (SAR) (Hirata and Shiozawa 2003, Morimoto *et al* 2016, Foster *et al* 2018) and absorbed or epithelial power density [hereafter referred to as absorbed power density (APD)] (Hashimoto *et al* 2017, Funahashi *et al* 2018). The current guidelines (IEEE C95.1 2019, ICNIRP 2020a) use the SAR averaged over 10 g of tissue (10 g SAR) for frequencies $\leqslant$6 GHz and the APD averaged over 4 cm^2 (4 cm^2 APD) for frequencies >6 GHz, as these metrics to correlate with the maximum temperature rise for local exposure. In computational assessments, anatomically realistic human body and head models are utilized (Hirata *et al* 2021). Although several groups have conducted computational studies to explore the rationale behind these guidelines, the majority have relied on a simplified model with layered structure (Gustrau and Bahr 2002, Sasaki *et al* 2017, Li *et al* 2019, Kapetanović and Poljak 2023) or a single head model (Bernardi *et al* 1998, Mcintosh and Anderson 2011, Kodera *et al* 2018a, Bonato *et al* 2021), with only three exceptions (Fujimoto *et al* 2006, Morimoto *et al* 2016, Morelli *et al* 2021).

The segmentation of medical images is commonly used to classify tissue types when constructing these models (Zaidi and Tsui 2009, Makarov *et al* 2017). These segmentation-based models are widely utilized for the safety assessment of electromagnetic field exposure (Lee *et al* 2002, Beard *et al* 2006), including the evaluation of temperature rise for RF exposure (Fujimoto *et al* 2006, Miura *et al* 2021, Kodera *et al* 2023a). In these models, each tissue is typically assigned a single set of dielectric and thermal properties. One of the significant sources of uncertainty in these computations is the variability of dielectric properties of tissues (Reilly and Hirata 2016, McCann *et al* 2019, Rashed *et al* 2020, Luo *et al* 2023). In particular, the dielectric properties change rapidly at the boundary between tissues with high and low water content (Sasaki *et al* 2022). In the research agenda of international guidelines, the necessity for further improvements in dosimetry methods is explicitly highlighted (ICNIRP 2025).

In (Rashed *et al* 2020), we have proposed a deep learning-based approach for human head modeling, where tissue dielectric properties and tissue density are directly estimated from magnetic resonance (MR) images. Comparison between this segmentation-free approach and conventional segmentation-based models have shown good agreement (within 5%). A notable advantage of this deep learning-based approach is the ability to reduce the computational cost, which is mainly used for anatomical segmentation, potentially allowing the assessment of individual variability in dosimetry, which could be valuable for discussing the safety factors. An open question remains regarding the impact of tissue thermal parameters on multiphysics dosimetry, specifically in the context of electromagnetic fields and thermodynamics. Due to individual water content variability, thermal parameters are also affected (Michel *et al* 2017). It would be beneficial to evaluate power absorption and temperature rise in such head models, considering the modeling approach and assessing its impact on previous findings.

In this study, we investigated the impact of segmentation-based versus segmentation-free methods for localized RF electromagnetic field exposure to clarify the computational uncertainty and variability in dose estimation caused by the human head modeling. To address these challenges, this study extended a deep learning-based approach to include the modeling of human head thermal parameters, where our previous studies considered only for dielectric properties. The main emphasis of this work is to quantify the differences introduced by conventional segmentation-based modeling compared to a segment free approach that incorporates smooth transitions in tissue properties.

2. Materials and methods

The computational methods for electromagnetics and thermodynamics in biological tissues were the same as in our previous studies (Hirata and Shiozawa 2003). Internal physical quantities in biological tissues are virtually impossible to validate compared to measurement. These computational methods were verified through several intercomparison studies led by the IEEE ICES Subcommittee 6 working groups (Diao *et al* 2021, Li *et al* 2021, 2023, 2024).

2.1. Human head model generation

To generate a realistic human head model using the segmentation-based and segmentation-free approaches, a set of T1- and T2-weighted MR images of 18 adult males (43.22 $\pm$ 9.77 years) was used. The MR images were obtained from the Brian Multimodality dataset. A set of 10 subjects of the 18 subjects were used for training the deep learning network and the remaining eight were used as test and validate the developed head models.

We consider two strategies; one considers the conventional method for human head model generation using anatomical segmentation of different tissues and set uniform dielectric properties to each segmented

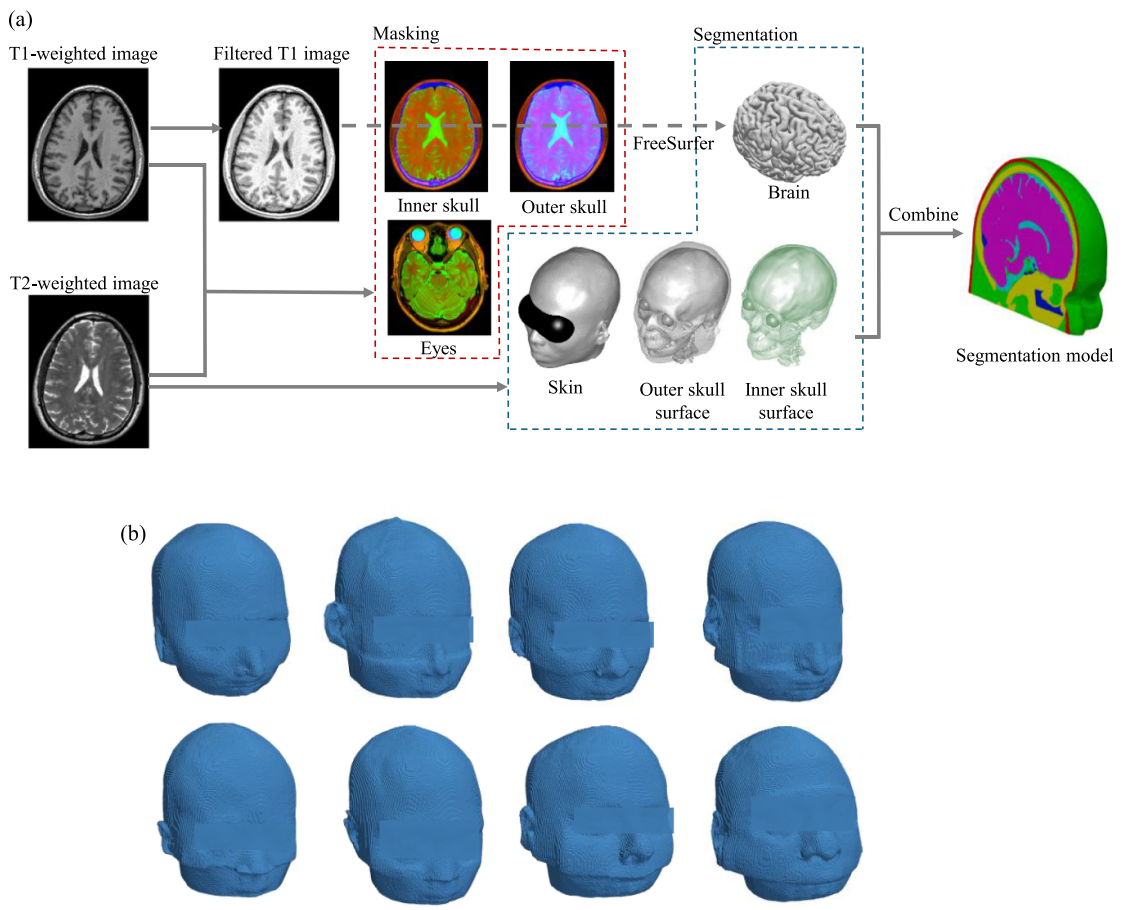

Figure 1. (a) Diagram of the segmentation-based head model construction from MR images and (b) surface view of the eight anatomical human head models.

volume. We refer to this strategy as 'segmentation-based' head model. The other is based on an automatic generation of dielectric properties using deep learning model directly from the MR images. We refer to this method as 'segmentation-free' head model. The segmentation-based models were segmented into 13 distinct tissues using a semi-automatic method (Laakso *et al* 2015) with a resolution of 0.5 × 0.5 × 0.5 mm. The human brain was specifically constructed from T1-weighted MR images using the FreeSurfer image analysis software version 5.3.0 (Fischl 2012). The remaining head parts were constructed from T1- and T2-weighted MR images and combined with the brain models with several manual-tuned parameters, as illustrated in figure 1. The dielectric and thermal properties were assigned as a unique values corresponding to each segmented tissue. The dielectric properties in each tissue were assigned using the 4-Cole–Cole dispersion model (Gabriel *et al* 1996b), and the thermal properties were the same as those used in our previous studies (Duck 1990, Hirata *et al* 2003). Many of the thermal parameters were derived considering the water content of each tissue (McIntosh and Anderson 2010).

In our previous study, we developed a deep learning network, CondNet, to generate a dataset of relative permittivity, conductivity, and tissue density without segmentation (Rashed *et al* 2020). A notable feature of this model is its capability to accommodate spatially smooth variations in dielectric properties, in contrast to conventional segmentation-based models, where uniform dielectric properties are assigned to distinct tissues. The network processes MR images (T1/T2-weighted) to generate the corresponding volume model of the dielectric properties determined based on the 4-Cole–Cole dispersion model (Gabriel *et al* 1996b). The segmentation-free models have demonstrated a superior representation of realistic simulation in several electromagnetic exposure studies (Diao *et al* 2022, 2024). In this study, this CondNet architecture was extended to generate a dataset of tissue thermal properties, including thermal conductivity, blood perfusion rate, specific heat, and mass per unit volume, which are needed to compute the temperature rise. A schematic overview of this procedure is presented in figure 2, which outlines the process for generating tissue thermal properties.

This approach processed T1- and T2-weighted MR slices as dual inputs and generated multiple outputs representing the distributions of specific heat, thermal conductivity, and blood perfusion rate. The key

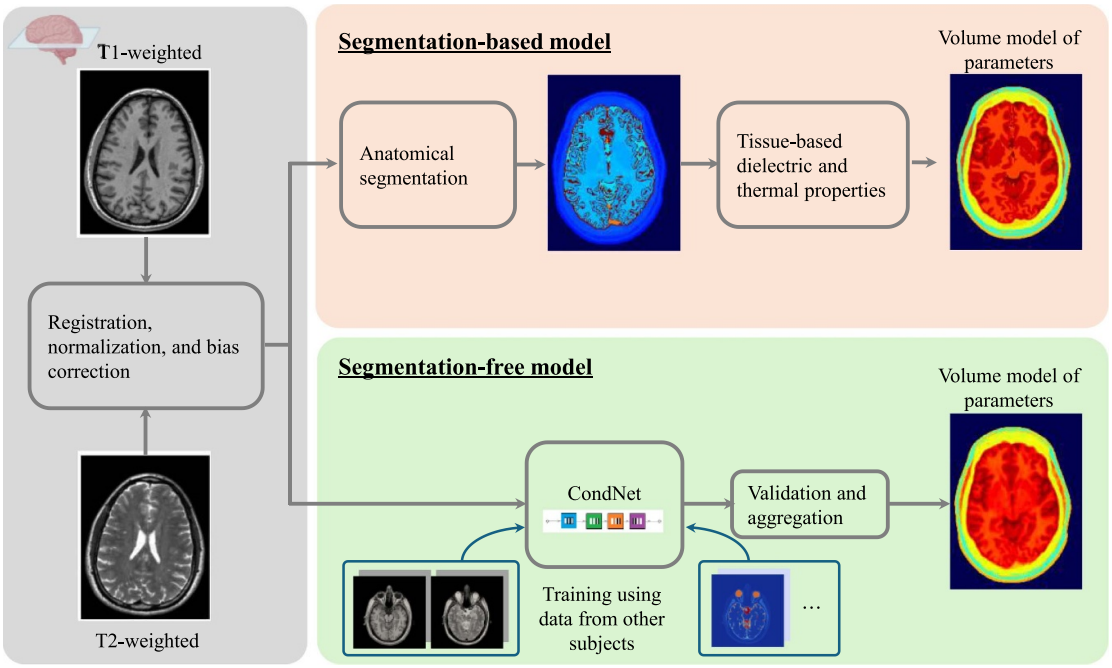

Figure 2. Schematic illustration of the generation procedure for a segmentation-free volume model of thermal parameters. The segmentation-free model is employed for computing the induced electric field and temperature rise.

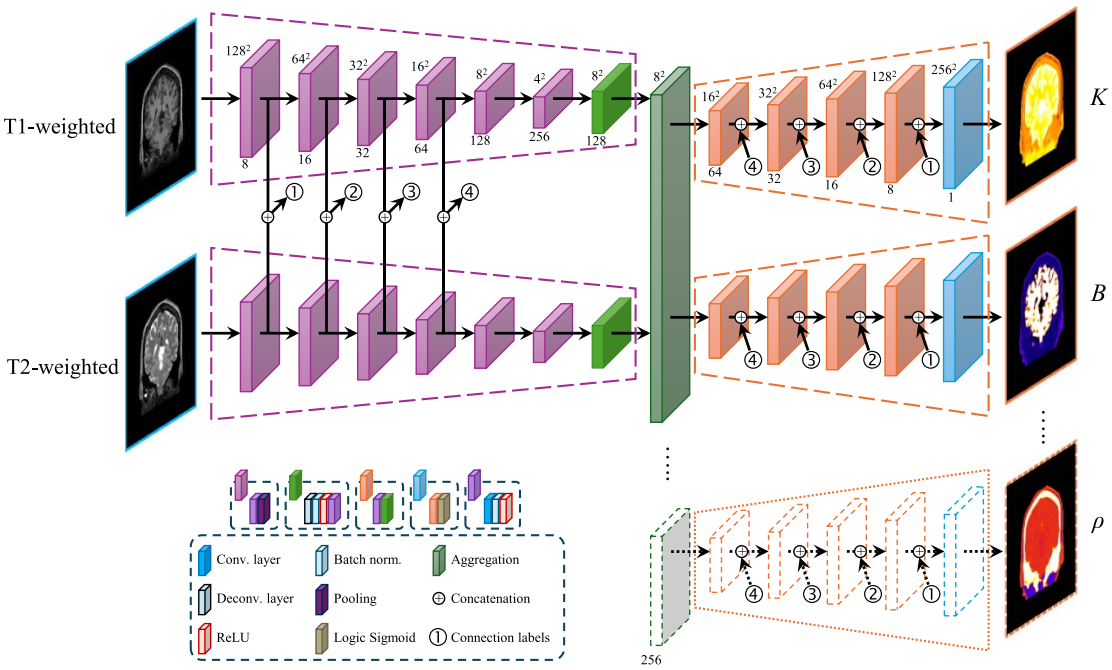

Figure 3. Overview of the framework for generating segmentation-free volume model of thermal parameters.

components of this architecture are the convolution and deconvolution operations, which simultaneously convert gray-scale values of MR images into the corresponding physical properties of tissues. The encoders composed of multiple convolution layers were designed to independently learn anatomical features, whereas the decoders were structured to map these features to various properties concurrently as shown in figure 3.

2.2. Electromagnetic computation

The finite-difference time-domain (FDTD) method (Taflove *et al* 2005) was employed to conduct electromagnetic dosimetry on human models exposed to RF fields. These grid resolutions adhered to less than one-tenth wavelength, with the accuracy of numerical dosimetry (Taflove *et al* 2005). A convolutional

Figure 4. Exposure scenarios.

perfectly matched layer (Roden and Gedney 2000) was employed to truncate the simulation domain, the exceptional absorption efficacy of which was corroborated for near-field and evanescent field scenarios. The SAR (IEEE C95.1 2019, ICNIRP 2020a) is defined as:

$$\mathrm{SAR}\,(x,y,z) = \frac{\sigma\,(x,y,z)}{2\rho\,(x,y,z)}|E\,(x,y,z)\,|^2 \tag{1}$$

where $E(x, y, z)$ represents the peak electric field at position (x, y, z), and σ and ρ are the conductivity and mass density of the tissue at position (x, y, z), respectively.

2.3. Temperature computation

The approach for computing the temperature rise in biological tissues was the same as in our previous studies (e.g. (Kodera and Hirata 2018, Kodera *et al* 2018b)). The temperature in the human model was followed in the time domain by solving a bioheat transfer equation (Pennes 1948). Excessive temperature rise caused thermoregulatory changes, such as vasodilation and sweating. However, the local temperature rise caused by the exposure below the limit at occupational exposure in the international guidelines was <1 °C–2 °C (Laakso and Hirata 2011); they were assumed to be negligible (Kodera *et al* 2018a).

A bioheat transfer equation (Pennes 1948), which incorporates the effect of blood perfusion, heat conduction, and electromagnetically induced heating SAR, can be written as follows:

$$\begin{aligned} C\,(x,y,z)\,\rho\,(x,y,z)\,\frac{\partial T\,(x,y,z,t)}{\partial t} &= \nabla\cdot\left(K\,(x,y,z)\,\nabla T\,(x,y,z,t)\right) + \rho\,(x,y,z)\,\mathrm{SAR}\,(x,y,z) \\ &\quad + M\,(x,y,z) - B\,(x,y,z)\left[T\,(x,y,z,t) - T_{\mathrm{B}}\,(x,y,z)\right], \end{aligned} \tag{2}$$

where C is the specific heat capacity, T is the tissue temperature, K is the tissue thermal conductivity, M is the metabolic heat generation, B is the term associated with blood flow, T_{B} is the blood temperature, and t is the time variable. The blood temperature was assumed to remain constant both spatially and temporally within the tissue, as the localized exposure scenarios were deemed insufficient to cause a core temperature rise (Kodera *et al* 2018a). Consequently, the blood temperature T_{B} in (2) was fixed at 37 °C.

The boundary condition for (2) is defined as

$$-K\,(x,y,z)\,\frac{\partial T\,(x,y,z,t)}{\partial n} = H\cdot\left(T_s\,(x,y,z,t) - T_e\,(t)\right), \tag{3}$$

where $\boldsymbol{n}$ denotes the normal component to the skin surface, and T_s, T_e, and H represent skin temperature, ambient temperature (which is position-independent), and the heat transfer coefficient, respectively. In the frequency range considered here (>6 GHz), heating was concentrated within a thin layer, typically only a few millimeters or less from the surface (Funahashi *et al* 2018, Ziskin *et al* 2018). Due to the significant temperature gradients near the surface, heat transport occurs primarily through thermal conduction.

2.4. Exposure scenarios

Half-wave dipole antennas operating at frequencies of 1, 3, 6, 10 GHz were considered as a canonical near-field source. The antenna was positioned 15 mm away from the scalp, near the temporal lobe, as illustrated in figure 4. To ensure resonance at the respective frequencies, the length of each dipole was adjusted to approximately half the wavelength of the corresponding RF field. Electromagnetic and thermal computations were conducted using parallel computation techniques to optimize processing time and enhance computational efficiency.

2.5. Post-processing

In the context of local exposure, the exposure guidelines utilized 10 g SAR and 4 cm^2APD as physical quantities to set the limit. These quantities are surrogate of the peak local and superficial temperature rises. The 10 g SAR at coordinates (x', y', z') with an averaging volume was determined according to the procedures outlined in IEEE C95.3 (2002):

$$\mathrm{SAR}_{10\mathrm{g}}(x',y',z') = \frac{\iiint_V \rho(x,y,z)\cdot \mathrm{SAR}(x,y,z)\,\mathrm{d}x\mathrm{d}y\mathrm{d}z}{\iiint_V \rho(x,y,z)\,\mathrm{d}x\mathrm{d}y\mathrm{d}z}, \tag{4}$$

where V is the volume of 10 g tissue in a cubic shape centered at (x', y', z') (IEEE-C95.3 2002).

The 4 cm^2 APD is as follows:

$$\mathrm{APD}_{4\mathrm{cm}^2}(y',z') = \frac{\iint_A \mathrm{d}y\mathrm{d}z \int_0^{x_{\max}} \rho(x,y,z)\cdot \mathrm{SAR}(x,y,z)\,\mathrm{d}x}{A}, \tag{5}$$

where the variable x represents the axis perpendicular to the surface. A represents the averaged area of 4 cm^2, centered at the coordinates (y', z'). APD was assessed by integrating along the wave propagation direction. As demonstrated in (Diao *et al* 2020), the differences in APD due to various averaging schemes for non-planar surface models are <20%.

The heating factor is defined as the ratio of the 'spatially peak temperature rise at the thermally steady state' to the 10 g SAR and the 4 cm^2 APD.

3. Results

3.1. Comparison of tissue thermal properties between segmentation-based and segmentation-free models

Figure 5 illustrates an example of the tissue thermal property distribution for a single subject, comparing the segmentation-based and segmentation-free head models. Although the distributions in both models are generally consistent, the segmentation-free model exhibits smoother transitions, particularly at tissue boundaries. A box plot comparing thermal properties and mass density in the segmentation-based and segmentation-free models is presented in figure 6. In figure 6, a good consistency is observed between the two models, especially when accounting for the variability in water content of human tissues. For all four physical parameters, the values in the segmentation-free models fall within the range of the standard deviation of segmentation-free model.

3.2. SAR and temperature distributions in different models at 1 and 10 GHz

The SAR distributions at 1 and 10 GHz are shown in figure 7. In figure 7, the SAR in the segmentation-free model is distributed more smoothly than in the segmentation-based model, even in different head models. This tendency is significant at lower frequencies where the penetration depth is large. Thermally steady-state temperature rise distributions at 1 and 10 GHz are shown in figure 8. In figure 8, the temperature rise is more smoothly distributed in the segmentation-free model; however, the difference in temperature distribution between the segmentation-based and segmentation-free models is less pronounced than that in the SAR distribution, due to the influence of thermal diffusion. The differences in the temperature rise distribution are observed around the location of the peak temperature rise.

3.3. Statistical evaluation

Figure 9(a) shows the peak 10 g SAR and the peak 4 cm^2 APD for segmentation-based and segmentation-free head models. Figure 9(a) shows the maximum discrepancy between the two models is 14.5%, with the mean values closely aligned. The relative standard deviation (RSD) of inter-subject variability for the peak 10 g SAR in the segmentation-based model was 15.2%, 21.1%, and 16.0% at 1 GHz, 3 GHz, and 6 GHz, respectively. In the segmentation-free model, the corresponding RSD values were 19.8%, 20.9%, and 9.4%. The RSD for 4 cm^2 APD at 10 GHz was 6.2% in the segmentation-based models and 5.0% in the segmentation-free models, respectively.

The difference in peak 10 g SAR and peak 4 cm^2 APD between segmentation-based and segmentation-free models in each subject is 5.4 ± 4.6% (mean ± SD) and 4.6 ± 3.3%, respectively, indicating strong consistency between the averaged SAR and APD across different head models. The variability in the spatial peak temperature rise is shown in figure 9(b). In figure 9(b), the difference in the maximum temperature rise between the segmentation-based and segmentation-free models is 7.6 ± 6.4%. Figures 9 (a) and (b) also indicate the variability among subjects is generally smaller in the segment-free models, except at

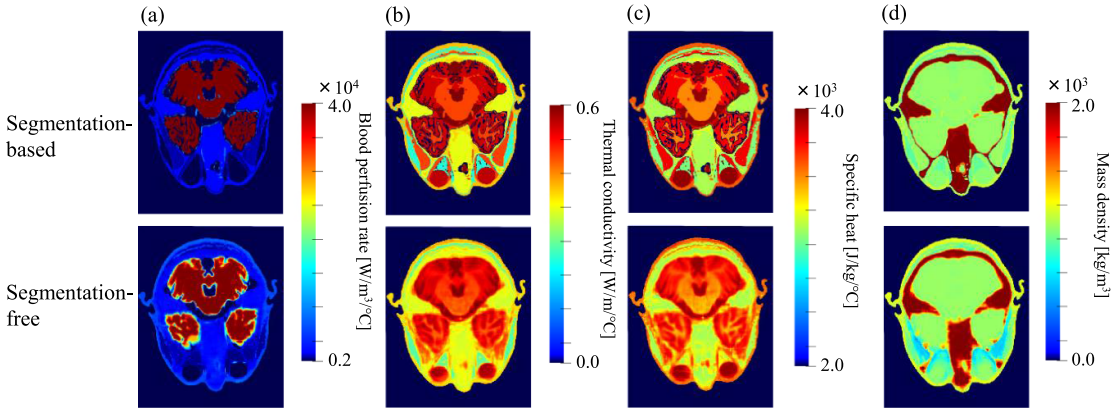

Figure 5. Distribution of thermal properties and mass density in segmentation-based and segmentation-free models: (a) blood perfusion rate, (b) thermal conductivity, (c) specific heat, and (d) mass density.

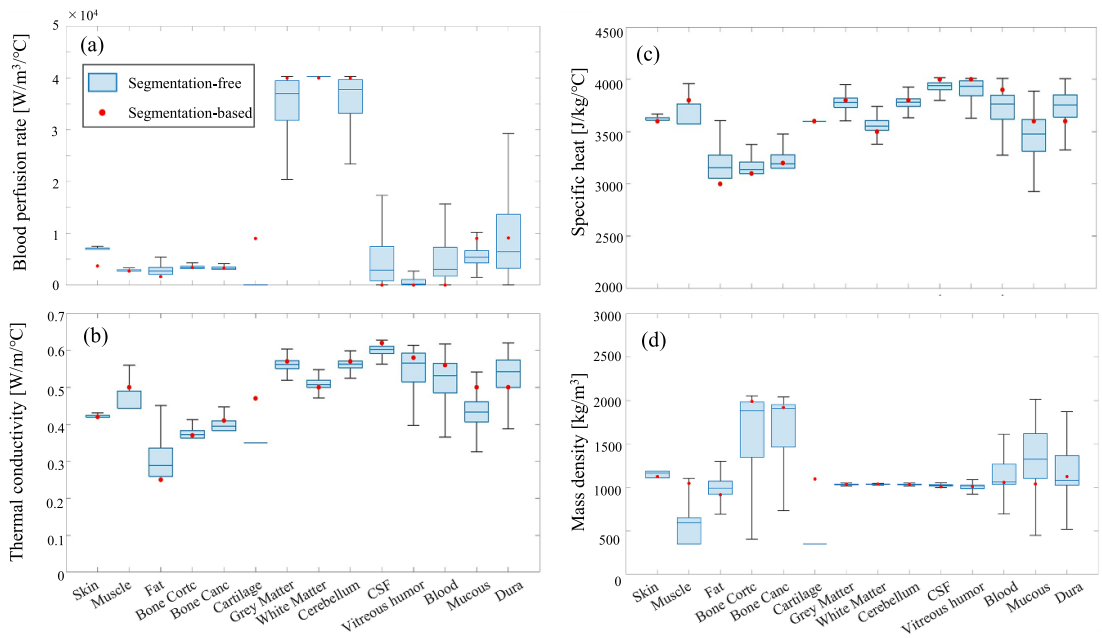

Figure 6. Comparison of thermal properties and mass density between the segmentation-based and segmentation-free models: (a) blood perfusion rate, (b) thermal conductivity, (c) specific heat, and (d) mass density.

1 GHz. Notably, the inter-subject variability inter-subject reduced above 6 GHz, with the RSD reducing by 7.1 and 3.4 percentage points at 6 and 10 GHz, respectively, compared to the segmentation-based model.

Figure 10 shows the variability in the heating factor. The difference in heating factor between the segmentation-based and segmentation-free models in each subject is 7.0 ± 5.1%. Their mean values are comparable, indicating good agreement in the heating coefficient between the two models despite their different physical properties. The maximum RSD in the inter-subject variability of heating factor was 15.0% in the segmentation-based models at 3 GHz, while the minimum RSD was 4.7% in the segmentation-free models at 6 GHz.

4. Discussion

In the development of RF exposure guidelines, the assessment of power absorption and temperature rise in the head models is essential. This study focused on improving the head modeling quality for evaluating temperature rise, which is essential to assess the RF exposure. A significant distinction from these earlier works is the consideration of inter-individual variability in electrical and thermal parameters, previously treated uniformly across tissues using deep-learning approach. The primary sources of dosimetric uncertainty when using voxel models include abrupt changes in physiological parameters at tissue interfaces and stair-casing errors at surface volume boundaries. Segmentation-free models help mitigate these

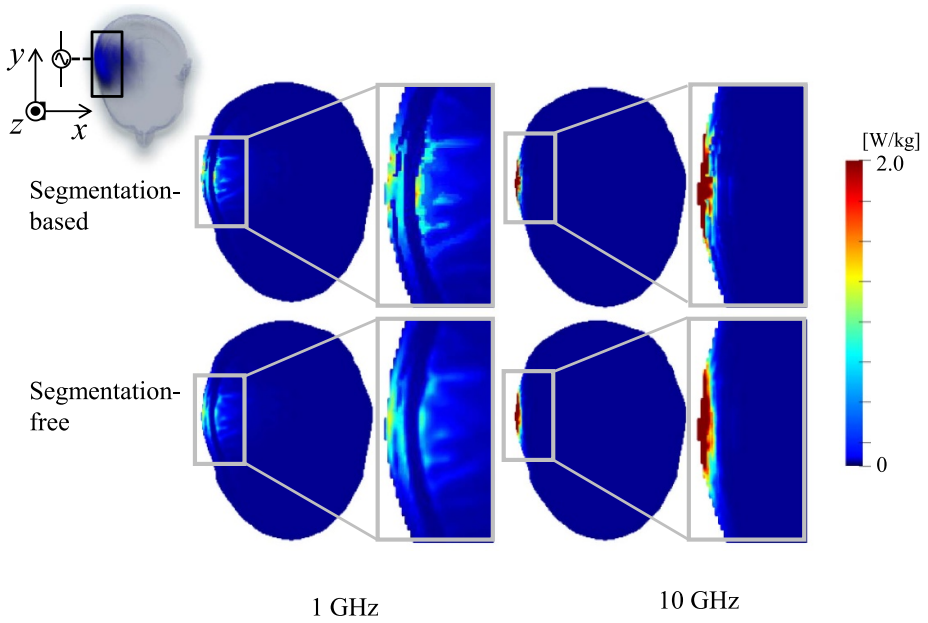

Figure 7. SAR distribution in a subject at 1 and 10 GHz. The antenna input power was set to 0.1 W.

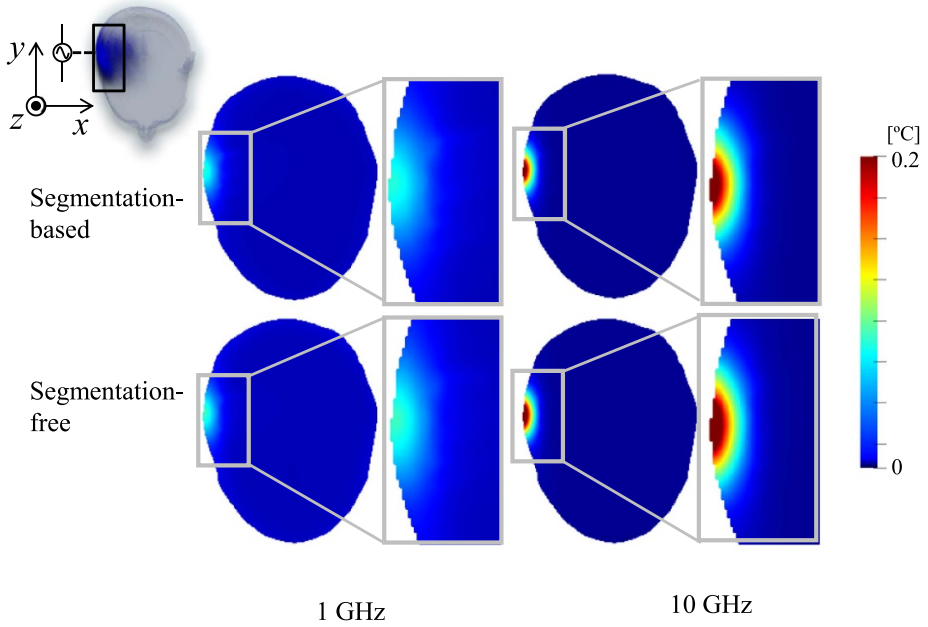

Figure 8. Distribution of thermally steady-state temperature rise distribution in a subject at 1 and 10 GHz. The antenna input power was set to 0.1 W.

stair-casing errors, particularly at tissue interfaces, by ensuring smoother transitions of physiological parameters between different tissues. Note that deep learning has often been applied to bioelectromagnetics to estimate the induced electric field or SAR from medical images (Bizopoulos and Koutsouris 2019, Yokota *et al* 2019, Zhang *et al* 2020, Gokyar *et al* 2021, Sathi *et al* 2021, Esmaeili *et al* 2023, Kapetanović and Poljak 2023, Esmaeili *et al* 2024, Khan *et al* 2024, 2019, Tur *et al* 2024). In contrast, this study focused on the non-uniform distribution of tissue thermal properties, estimated directly from medical images.

This study evaluated the variability of SAR, APD and temperature rise for local exposure from a dipole antenna using two distinct head models: segmentation-free and segmentation models. The primary difference between these models was in the inclusion or absence of tissue segmentation during model generation. Eight individual head models were constructed from MR images, whereas most previous studies for evaluating microwave-induced temperature typically used a few head models (Karampatzakis and Samaras 2013, Bonato *et al* 2021, Morelli *et al* 2021, Kapetanović and Poljak 2023, Haider *et al* 2023).

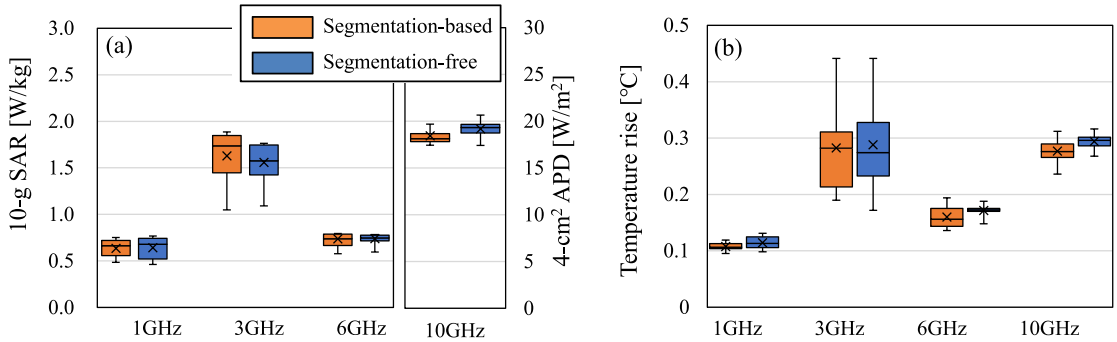

Figure 9. Inter-subject variability of (a) 10 g SAR and 4 cm^2 APD, and (b) temperature rise in the segmentation-based and segmentation-free models. The antenna input power was set to 0.1 W.

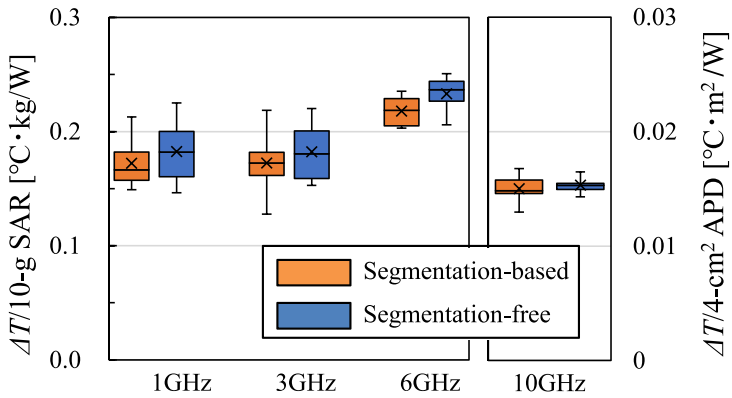

Figure 10. Inter-subject variability of the heating factor in the segmentation-based and segmentation-free models.

Computational results show a good agreement between the two different types of head models. One notable feature of the segmentation-free model is that, unlike the conventional models with tissue segmentation, thermal constants vary even within the same tissue. The thermal constants in the segmentation-based models generally fall within the variability range observed in the segmentation-free model. This tendency is consistent with dielectric properties in the segmentation-free model (Rashed *et al* 2020) and confirmed for all four thermal properties. One potential explanation for internal variability of the tissue is that the water contents of white matter in adults range from 68% to 77% (ICRP 2002). Similarly, a variability of thermal constant for different head tissues was presented in (Duck 1990, McIntosh and Anderson 2010). The segmentation-free model captures this variability, whereas segmentation models apply a uniform thermal constant to each tissue.

The computational results showed that the spatial distribution of SAR and temperature rise were similar in the segmentation-based and segmentation-free models, and with the resulting 10 g SAR and 4 cm^2 APD generally in good agreement. Simple spatial averaging of the SAR distribution from a conventional segmentation-based model cannot fully replicate the results of the segmentation-free approach. This is evident from the fact that the spatially averaged values, 10 g SAR and 4 cm^2 APD, did not perfectly match between both models. One possible explanation is that deep learning-based methods inherently capture variations in tissue properties within each voxel, rather than assigning uniform characteristics to segmented tissues.

Regarding inter-subject variability, the maximum RSD was observed at 3 GHz for both the basic restriction and temperature rise, with a general decreasing trend as frequency increased. Variability among the subjects was generally smaller in the segmentation-free models. In particular, the inter-subject variability of 10 g SAR was reduced in the segmentation-free head models, except at 1 GHz. Additionally, the RSD was lower in the segmentation-free model compared to the segmentation-based model at 10 GHz. The inter-subject variability dominated the variability due to differences between the two methods. This suggests that the voxel-wise heterogeneous assignment of dielectric and thermal properties in the segmentation-free models smooths inter-tissue variations, leading to less pronounced individual differences than in the segmentation models. This finding underscores the potential advantage of segmentation-free models in

minimizing inter-subject variability, a critical consideration for personalized exposure assessment (e.g. setting an appropriate reduction factor when deriving the limits) and personalized medicine. At 1 GHz, the segmentation-free model had higher inter-subject variability. This is because, at lower frequencies, RF energy penetrates deeper into tissues, where subtle variations in dielectric and thermal properties have a greater cumulative impact. As shown in figure 6, deeper tissues exhibit greater variability in thermal properties compared to superficial tissues. Consequently, inter-subject variability may increase at lower frequencies.

As noted in the guidelines (ICNIRP 2020a), the initial temperature rise in tissues is primarily governed by specific heat capacity, whereas the steady-state temperature rise is predominantly influenced by blood perfusion rate (also see (Bernardi *et al* 2000, Hirata *et al* 2006)). As shown in figure 6(a), a low skin blood perfusion rate was selected among measured values in the segmentation-based models for a conservative estimation. In contrast, the blood perfusion rate in skin, as estimated by MR image, is approximately twice as high, which is close to the measured value observed in regions with high blood flow, such as the head (Utsunomiya *et al* 1990, Foster *et al* 2016). If these higher values were applied, the heating factor reported here could be lower. The effect on surface temperature rise was estimated at ∼15% at frequencies >6 GHz (Laakso *et al* 2017, Sasaki *et al* 2017). However, the blood perfusion rate highly depends on individuals and the ambient conditions; these remain future topics.

Furthermore, this study incorporated eight individual head models, considering anatomical differences, such as skin and skull thickness variations. Often overlooked in simpler models, these factors play a role in determining SAR and temperature rise for RF exposure. At (quasi-) millimeter waves, the slab model has been used rather than realistic anatomical shapes due to the high computational costs involved, as discussed extensively in (Colella *et al* 2023). Even at 10 GHz, the segmentation-free models demonstrated less variability of SAR and temperature rise compared to their segmented counterparts, suggesting the robustness of segmentation-free models at higher frequencies.

Another issue is accurately modeling tissue thickness (Rashed *et al* 2019), particularly for skin layers. Most anatomical models used in previous studies had a relatively coarse resolution of a few millimeters. As a result, tissue thickness in high-resolution models reconstructed by these models is constrained to be an integer multiple of the original resolution. However, the actual thickness of human skin varies depending on the specific location of the body and individual anatomical differences. These variations are particularly important at frequencies <20 GHz, where power absorption is more influenced by the thickness of superficial layers as the electromagnetic field penetrates deeper into the tissue (Taguchi *et al* 2022, Kodera *et al* 2023b). In this study, the head models with a resolution of 0.5 mm were constructed directly from MR image, allowing for a more accurate representation of fine anatomical details, such as the various skin thicknesses.

In the ICNIRP guidelines (ICNIRP 2020a), the basic restriction (exposure limit inside the body) of 10 g SAR for occupational exposure was set at 10 W kg^{-1} to ensure that the tissue temperature rise for local exposure remains <2.5 °C in superficial tissues and <1 °C in the brain and cataract, and other sensitive tissues. These values were derived by applying reduction factors to the 'operational thresholds'. The include biological variability within the population (e.g. age, sex), differences in baseline conditions (e.g. tissue temperature), environmental influences (e.g. air temperature, humidity, clothing), dosimetric uncertainties in deriving exposure values, uncertainties in health-related research, and general conservative considerations (ICNIRP 2020a). Our results showed that the heating factor in the skin of the segmentation-free models was in the range of 0.17 °C–0.24 °C kg W^{-1}, consistent with previous studies (Van Leeuwen *et al* 1999, Bernardi *et al* 2000, Gandhi *et al* 2001), which were in the range of 0.11 °C–0.27 °C kg W^{-1}.

Above 6 GHz, the temperature rise at 100 W cm^{-2} of 4 cm^2 APD corresponds to the skin temperature rise limit should be <2.5 °C. Our results confirmed that the temperature rise at 100 W cm^{-2} of APD was 1.5 °C which is well below 2.5 °C. It is worth nothing that the heating factor for APD is somewhat reduced at 10 GHz, because superficial heating becomes more dominant at frequencies of >30 GHz.

The limitations of this study are as follows. The segmentation-free models used are based on the tissue property databases of Duck (1990) and Gabriel *et al* (1996a), which are widely utilized in bioelectromagnetics, especially computational dosimetry studies. However, these databases contain *in-vitro* measurements and extrapolated data from animal studies, and improvements to the databases may provide further robustness to the architecture. In further advances in human exposure assessment, it is important to consider age, race, and gender differences (ICNIRP 2020a). The age-dependency of electrical and thermal properties of tissues were reported in (Schmid and Überbacher 2005, Peyman *et al* 2007), but only for a limited number of tissues, and a comprehensive database that covers their variations is not yet available. ICNRP also mentioned that '*Further exploration of the effects of age, sex, and other morphological differences on these models and their outcomes would be of great benefit for the derivation of reduction factors in future guidelines*' (ICNIRP 2025). If limited data on physical properties related to age, sex, and race were available, the architecture in this study could implicitly and comprehensively incorporate changes in physical characteristics related to age, sex, and race. However, this feature has not been explicitly verified, and further

research is needed to confirm and refine the model's ability to comprehensively address these variations. These limitations highlight the need for future studies to develop a more robust database and validate model performance across a range of demographic and biological factors.

5. Conclusion

This study explored the effectiveness of segmentation-free modeling techniques in assessing human head exposure to localized RF electromagnetic fields. By comparing the segmentation-based model with the segmentation-free model, the SAR, APD, and temperature rise agreed within 8% on average. Notably, the segmentation-free approach exhibited less variability among subjects, particularly evident at higher frequencies where superficial heating prevails. These findings underscore the potential of deep learning-based head modeling for enhancing accuracy and efficiency in RF dosimetry, offering valuable implications for individual variability assessment and adherence to international guidelines.

Data availability statement

The data cannot be made publicly available upon publication because no suitable repository exists for hosting data in this field of study. The data that support the findings of this study are available upon reasonable request from the authors.

Acknowledgment

This work was supported by JSPS KAKENHI Grant Nos JP24K00868, JP21K04000 and JP22K12765. This work was supported by 'Joint Usage/Research Center for Interdisciplinary Large-scale Information Infrastructures (JHPCN)' in Japan (Project ID: jh240009).

During the preparation of this paper, the authors partly used ChatGPT-4.0 to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

ORCID iDs

Sachiko Kodera https://orcid.org/0000-0001-6595-0742
Essam A Rashed https://orcid.org/0000-0001-6571-9807
Yinliang Diao https://orcid.org/0000-0002-6492-4515
Hiroyuki Takizawa https://orcid.org/0000-0003-2858-3140
Akimasa Hirata https://orcid.org/0000-0001-8336-1140

References

Beard B B *et al* 2006 Comparisons of computed mobile phone induced SAR in the SAM phantom to that in anatomically correct models of the human head *IEEE Trans. Electromagn. Compat.* **48** 397–407

Bernardi P, Cavagnaro M, Pisa S and Piuzzi E 1998 SAR distribution and temperature increase in an anatomical model of the human eye exposed to the field radiated by the user antenna in a wireless LAN *IEEE Trans. Microw. Theory Tech.* **46** 2074–82

Bernardi P, Cavagnaro M, Pisa S and Piuzzi E 2000 Specific absorption rate and temperature increases in the head of a cellular-phone user *IEEE Trans. Microw. Theory Tech.* **48** 1118–26

Bizopoulos P and Koutsouris D 2019 Deep learning in cardiology *IEEE Rev. Biomed. Eng.* **12** 168–93

Bonato M, Dossi L, Chiaramello E, Fiocchi S, Gallucci S, Tognola G, Ravazzani P and Parazzini M 2021 Human RF-EMF exposure assessment due to access point in incoming 5G indoor scenario *IEEE J. Electromagn. RF Microw. Med. Biol.* **5** 269–76

Colella M, Di Meo S, Liberti M, Pasian M and Apollonio F 2023 Advantages and disadvantages of computational dosimetry strategies in the low mmW range: comparison between multilayer slab and anthropomorphic models *IEEE Trans. Microw. Theory Tech.* **71** 4533–45

Diao Y, Li K, Sasaki K, Kodera S, Laakso I, El Hajj W and Hirata A 2021 Effect of incidence angle on the spatial-average of incident power density definition to correlate skin temperature rise for millimeter wave exposures *IEEE Trans. Electromagn. Compat.* **63** 1709–16

Diao Y, Rashed E A, Giaccone L, Laakso I, Li C, Scorretti R and Hirata A 2024 Effect of the conductivity variations on computed electric field induced in learning-based models *IEEE Access* **12** 188454–64

Diao Y, Rashed E A and Hirata A 2020 Assessment of absorbed power density and temperature rise for nonplanar body model under electromagnetic exposure above 6 GHz *Phys. Med. Biol.* **65** 224001

Diao Y, Rashed E A and Hirata A 2022 Induced electric field in learning-based head models with smooth conductivity for exposure to uniform low-frequency magnetic fields *IEEE Trans. Electromagn. Compat.* **64** 1969–77

Duck F A 1990 *Physical Properties of Tissues: A Comprehensive Reference Book* (Academic)

Esmaeili H, Yang C and Schuster C 2023 Physics inspired artificial neural network adaptation for SAR prediction in bio-EM problems *2023 IEEE MTT-S Int. Microwave Biomedical Conf. (Imbioc)* (IEEE) pp 1–3

Esmaeili H, Yang C and Schuster C 2024 Efficient iterative data generation using evaluation of prioritized input parameters in ANNs for SAR prediction in human head models at 13.56 MHz *IEEE Trans. Electromagn. Compat.* pp 1–11

Fischl B 2012 FreeSurfer *Neuroimage* **62** 774–81
Foster K R, Ziskin M C and Balzano Q 2016 Thermal response of human skin to microwave energy: a critical review *Health Phys.* **111** 528–41
Foster K R, Ziskin M C, Balzano Q and Bit-Babik G 2018 Modeling tissue heating from exposure to radiofrequency energy and relevance of tissue heating to exposure limits: heating factor *Health Phys.* **115** 295–307
Fujimoto M, Hirata A, Wang J, Fujiwara O and Shiozawa T 2006 FDTD-derived correlation of maximum temperature increase and peak SAR in child and adult head models due to dipole antenna *IEEE Trans. Electromagn. Compat.* **48** 240–7
Funahashi D, Hirata A, Kodera S and Foster K R 2018 Area-averaged transmitted power density at skin surface as metric to estimate surface temperature elevation *IEEE Access* **6** 77665–74
Gabriel C, Gabriel S and Corthout E 1996a The dielectric properties of biological tissues: I. Literature survey *Phys. Med. Biol.* **41** 2231
Gabriel S, Lau R W and Gabriel C 1996b The dielectric properties of biological tissues: III. Parametric models for the dielectric spectrum of tissues *Phys. Med. Biol.* **41** 2271–93
Gandhi O P, Qing-Xiang L and Kang G 2001 Temperature rise for the human head for cellular telephones and for peak SARs prescribed in safety guidelines *IEEE Trans. Microw. Theory Tech.* **49** 1607–13
Gokyar S, Robb F J L, Kainz W, Chaudhari A and Winkler S A 2021 MRSaiFE: an AI-based approach towards the real-time prediction of specific absorption rate *IEEE Access* **9** 140824–34
Gustrau F and Bahr A 2002 W-band investigation of material parameters, SAR distribution, and thermal response in human tissue *IEEE Trans. Microw. Theory Tech.* **50** 2393–400
Haider Z, Modolo J, Liberti M, Apollonio F and Zhadobov M 2023 Millimeter-wave induced heating of cutaneous nerves and capillaries *IEEE J. Microw.* **3** 170–80
Hashimoto Y, Hirata A, Morimoto R, Aonuma S, Laakso I, Jokela K and Foster K R 2017 On the averaging area for incident power density for human exposure limits at frequencies over 6 GHz *Phys. Med. Biol.* **62** 3124–38
Hirata A *et al* 2021 Assessment of human exposure to electromagnetic fields: review and future directions *IEEE Trans. Electromagn. Compat.* **63** 1619–30
Hirata A, Fujiwara O and Shiozawa T 2006 Correlation between peak spatial-average SAR and temperature increase due to antennas attached to human trunk *IEEE Trans. Biomed. Eng.* **53** 1658–64
Hirata A, Funahashi D and Kodera S 2019 Setting exposure guidelines and product safety standards for radio-frequency exposure at frequencies above 6 GHz: brief review *Ann. Des. Telecommun. Telecommun.* **74** 17–24
Hirata A and Kodera S 2020 Difference of ICNIRP guidelines and IEEE C95.1 standard for human protection from radio-frequency exposures *2020 Int. Symp. on Electromagnetic Compatibility—EMC EUROPE* (IEEE) pp 1–5
Hirata A, Morita M and Shiozawa T 2003 Temperature increase in the human head due to a dipole antenna at microwave frequencies *IEEE Trans. Electromagn. Compat.* **45** 109–16
Hirata A and Shiozawa T 2003 Correlation of maximum temperature increase and peak SAR in the human head due to handset antennas *IEEE Trans. Microw. Theory Tech.* **51** 1834–41
ICNIRP 2010 Guidelines for limiting exposure to time-varying electric and magnetic fields (1 Hz to 100 kHz *Health Phys.* **99** 818–36
ICNIRP 2020a Guidelines for limiting exposure to electromagnetic fields (100 kHz to 300 GHz) *Health Phys.* **118** 483–524
ICNIRP 2020b Principles for non-ionizing radiation protection *Health Phys.* **118** 477–82
ICNIRP 2025 Gaps in knowledge relevant to the "ICNIRP guidelines for limiting exposure to time-varying electric, magnetic and electromagnetic fields (100 kHz TO 300 GHz) *Health Phys.* **128** 190–202
ICRP 2002 *Basic Anatomical and Physiological Data for Use in Radiological Protection: Reference Values. A Report of Age- and Gender-Related Differences in the Anatomical and Physiological Characteristics of Reference Individuals*
IEEE C95.1 2019 IEEE standard for safety levels with respect to human exposure to electric, magnetic and electromagnetic fields, 0 Hz to 300 GHz *IEEE Std*
IEEE-C95.3 2002 IEEE recommended practice for measurements and computations of radio frequency electromagnetic fields with respect to human exposure to such fields, 100 kHz-300 GHz *IEEE* vol 126
Kapetanović A and Poljak D 2023 Machine learning-assisted antenna modelling for realistic assessment of incident power density on non-planar surfaces above 6 GHz *Radiat. Prot. Dosim.* **199** 826–34
Karampatzakis A and Samaras T 2013 Numerical modeling of heat and mass transfer in the human eye under millimeter wave exposure *Bioelectromagnetics* **34** 291–9
Khan A, Ghorbanian V and Lowther D 2019 Deep learning for magnetic field estimation *IEEE Trans. Magn.* **55** 1–4
Khan M R, Zekios C L, Bhardwaj S and Georgakopoulos S V 2024 A deep learning convolutional neural network for antenna near-field prediction and surrogate modeling *IEEE Access* **12** 39737–47
Kodera S, Gomez-Tames J and Hirata A 2018a Temperature elevation in the human brain and skin with thermoregulation during exposure to RF energy *Biomed. Eng. Online* **17** 1–17
Kodera S and Hirata A 2018 Comparison of thermal response for RF exposure in human and rat models *Int. J. Environ. Res. Public Health* **15** 2320
Kodera S, Hirata A, Funahashi D, Watanabe S, Jokela K and Croft R J 2018b Temperature rise for brief radio-frequency exposure below 6 GHz *IEEE Access* **6** 65737–46
Kodera S, Miura N, Diao Y, Inoue M, Hikage T, Taguchi K, Masuda H and Hirata A 2023a Whole-body exposure system using horn antennas with dielectric lens at 28 GHz *IEEE J. Electromagn. RF Microw. Med. Biol* **7** 65–72
Kodera S, Taguchi K, Diao Y, Kashiwa T and Hirata A 2023b Computation of whole-body average SAR in realistic human models from 1 to 100 GHz *IEEE Trans. Microw. Theory Tech.* **72** 91–100
Laakso I and Hirata A 2011 Dominant factors affecting temperature rise in simulations of human thermoregulation during RF exposure *Phys. Med. Biol.* **56** 7449–71
Laakso I, Morimoto R, Hirata A and Onishi T 2017 Computational dosimetry of the human head exposed to near-field microwaves using measured blood flow *IEEE Trans. Electromagn. Compat.* **59** 739–46
Laakso I, Tanaka S, Koyama S, De Santis V and Hirata A 2015 Inter-subject variability in electric fields of motor cortical tDCS *Brain Stimul.* **8** 906–13
Lee A K, Choi H D, Lee H S and Pack J K 2002 Human head size and SAR characteristics for handset exposure *ETRI J.* **24** 176–9
Li K, Diao Y, Sasaki K, Prokop A, Poljak D, Doric V, Xi J, Kodera S, Hirata A and El Hajj W 2021 Intercomparison of calculated incident power density and temperature rise for exposure from different antennas at 10-90 GHz *IEEE Access* **9** 151654–66
Li K *et al* 2023 Calculated epithelial/absorbed power density for exposure from antennas at 10–90 GHz: intercomparison study using a planar skin model *IEEE Access* **11** 7420–35

Li K *et al* 2024 Spatially averaged epithelial/absorbed power density for nonplanar skin models exposed to antenna at 10-90 GHz *IEEE Access* **12** 15379–89

Li K, Sasaki K, Watanabe S and Shirai H 2019 Relationship between power density and surface temperature elevation for human skin exposure to electromagnetic waves with oblique incidence angle from 6 GHz to 1 THz *Phys. Med. Biol.* **64** 065016

Luo Z, Li C, Liu D, Wang B, Zhang L, Ma Y, Xu K and Wang X 2023 Quantitative reconstruction of dielectric properties based on deep-learning-enabled microwave-induced thermoacoustic tomography *IEEE Trans. Microw. Theory Tech.* **71** 2652–63

Makarov S N, Noetscher G M, Yanamadala J, Piazza M W, Louie S, Prokop A, Nazarian A and Nummenmaa A 2017 Virtual human models for electromagnetic studies and their applications *IEEE Rev. Biomed. Eng.* **10** 95–121

McCann H, Pisano G and Beltrachini L 2019 *Variation in Reported Human Head Tissue Electrical Conductivity Values* vol 32 (Springer US)

Mcintosh R L and Anderson V 2011 SAR versus VAR, and the size and shape that provide the most appropriate RF exposure metric in the range of 0.5-6GHz *Bioelectromagnetics* **32** 312–21

McIntosh R L and Anderson V 2010 A comprehensive tissue properties database provided for the thermal assessment of a human at rest *Biophys. Rev. Lett.* **05** 129–51

Michel E, Hernandez D and Lee S Y 2017 Electrical conductivity and permittivity maps of brain tissues derived from water content based on T1-weighted acquisition *Magn. Reson. Med.* **77** 1094–103

Miura N, Kodera S, Diao Y, Higashiyama J, Suzuki Y and Hirata A 2021 Power absorption and skin temperature rise from simultaneous near-field exposure at 2 and 28 GHz *IEEE Access* **9** 152140–9

Morelli M S, Gallucci S, Siervo B and Hartwig V 2021 Numerical analysis of electromagnetic field exposure from 5G mobile communications at 28 GHZ in adults and children users for real-world exposure scenarios *Int. J. Environ. Res. Public Health* **18** 1–13

Morimoto R, Laakso I, De Santis V and Hirata A 2016 Relationship between peak spatial-averaged specific absorption rate and peak temperature elevation in human head in frequency range of 1-30 GHz *Phys. Med. Biol.* **61** 5406–25

Pennes H H 1948 Analysis of tissue and arterial blood temperatures in the resting human forearm *J. Appl. Physiol.* **1** 93–122

Peyman A, Holden S J, Watts S, Perrott R and Gabriel C 2007 Dielectric properties of porcine cerebrospinal tissues at microwave frequencies: *in vivo, in vitro* and systematic variation with age *Phys. Med. Biol.* **52** 2229–45

Rashed E A, Diao Y and Hirata A 2020 Learning-based estimation of dielectric properties and tissue density in head models for personalized radio-frequency dosimetry *Phys. Med. Biol.* **65** 065001

Rashed E A, Gomez-tames J and Hirata A 2019 Human head skin thickness modeling for electromagnetic dosimetry *IEEE Access* **7** 46176–86

Reilly J P and Hirata A 2016 Low-frequency electrical dosimetry: research agenda of the IEEE international committee on electromagnetic safety *Phys. Med. Biol.* **61** R138–R49

Roden J A and Gedney S D 2000 Convolution PML (CPML): an efficient FDTD implementation of the CFS-PML for arbitrary media *Microw. Opt. Technol. Lett.* **27** 334–9

Sasaki K, Mizuno M, Wake K and Watanabe S 2017 Monte Carlo simulations of skin exposure to electromagnetic field from 10 GHz to 1 THz *Phys. Med. Biol.* **62** 6993–7010

Sasaki K, Porter E, Rashed E A, Farrugia L and Schmid G 2022 Measurement and image-based estimation of dielectric properties of biological tissues—past, present, and future— *Phys. Med. Biol.* **67** 14TR01

Sathi K A, Hossain M A, Hosain M K, Hai N H and Hossain M A 2021 A deep neural network model for predicting electric fields induced by transcranial magnetic stimulation coil *IEEE Access* **9** 128381–92

Schmid G and Überbacher R 2005 Age dependence of dielectric properties of bovine brain and ocular tissues in the frequency range of 400 MHz to 18 GHz *Phys. Med. Biol.* **50** 4711–20

Taflove A, Hagness S C and Piket-May M 2005 *Computational Electrodynamics : the Finite-Difference Time-Domain Method* (The Electrical Engineering Handbook) vol 3 pp 629–70

Taguchi K, Kodera S, Hirata A and Kashiwa T 2022 Computation of absorbed power densities in high-resolution head models by considering skin thickness in quasi-millimeter and millimeter wave bands *IEEE J. Electromagn. RF Microw. Med. Biol* **6** 516–23

Tur Y, Vu J, Waktola S, Medetalibeyoglu A, Golestanirad L and Bagci U 2024 Prediction of MRI-induced power absorption in patients with DBS leads *2024 IEEE 37th Int. Symp. on Computer-Based Medical Systems (CBMS)* (IEEE) pp 490–5

Utsunomiya M, Ohura H, Takiwaki H, Arase S and ShigemiI F 1990 Measurement of skin blood flow by the laser Doppler flowmetry *Nishi Nihon Hifuka* **52** 1164–71

Van Leeuwen G M J, Lagendijk J J W, Van Leersum B J A M, Zwamborn A P M, Hornsleth S N and Kotte A N T J 1999 Calculation of change in brain temperatures due to exposure to a mobile phone *Phys. Med. Biol.* **44** 2367–79

Yokota T, Maki T, Nagata T, Murakami T, Ugawa Y, Laakso I, Hirata A and Hontani H 2019 Real-time estimation of electric fields induced by transcranial magnetic stimulation with deep neural networks *Brain Stimul.* **12** 1500–7

Zaidi H and Tsui B M W 2009 Review of computational anthropomorphic anatomical and physiological models *Proc. IEEE* **97** 1938–53

Zhang J, Zheng G, Krikidis I and Zhang R 2020 Fast specific absorption rate aware beamforming for downlink SWIPT via deep learning *IEEE Trans. Veh. Technol.* **69** 16178–82

Ziskin M C, Alekseev S I, Foster K R and Balzano Q 2018 Tissue models for RF exposure evaluation at frequencies above 6 GHz *Bioelectromagnetics* **39** 173–89